3W Well Data Viewer

A desktop GUI to inspect the real instances of the 3W dataset

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Overlapping samples

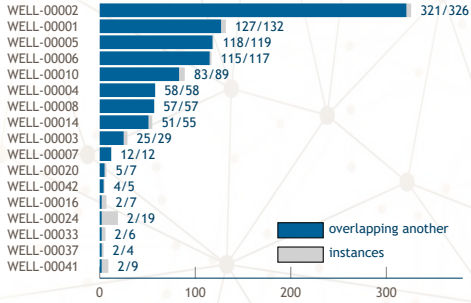
Timelines page ► How are the instances of one well distributed in time?

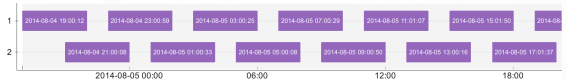
17
of 40 wells have overlapping instances

986
of 1,119 instances overlap another of their well

0
label collisions in the 912 overlapping pairs

Instances of the wells with overlaps (the other 23 have none)





Instances overlapping in WELL-00004.

Worries:

- Overlapping samples may land in **train and test**, causing leakage.
- Duplicated stretches pull **clusters** tighter, inflating scores.

Thankfully, 3W 2.0.0 has no label collisions, making joins principled and avoiding leakage.
Clustering must still be carefully calculated!

What a joined history is labeled as

Timelines page ▶ How are the instances of one well distributed in time?

12.3 % ▷ 2.9 %

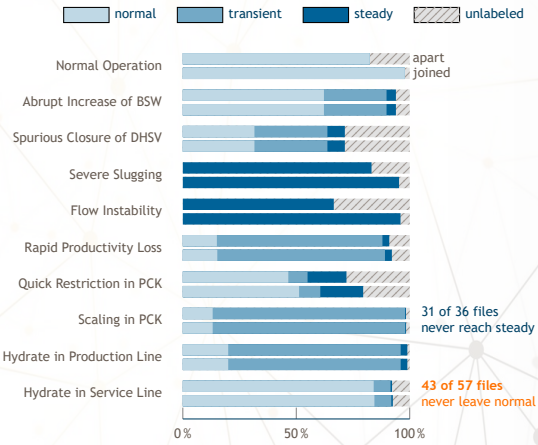
of unlabels after join

After the join, most unlabeled stretches appear trivial and might be recoverable.

Unlabeled stretch position	*n*	Duration	Share
head of a recording	208	1 h each	86.5 %
seam between equal labels	882	median 76 s, max 56 min	11.5 %
between two labels	16	median 6 min, max 1 h	2.0 %

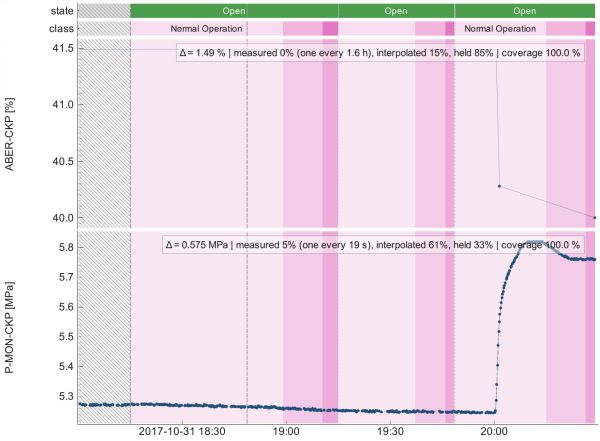
Appendix

Each fault folder's recorded time, by label, its files apart and joined



Histories whose labels do not add up

Timelines page ► How are the instances of one well distributed in time?



WELL-00004's joined Quick Restriction in PCK instances.

- Joined histories go back to normal **13 times** in 7 wells, **every time at a file boundary**.
- **WELL-00004**, left: three quick restrictions in 75 min, each back to normal 5 to 15 s after its steady state.
- **WELL-00006**: Abrupt Increase of BSW, steady, then normal 30 min later in a Scaling in PCK file. A water cut does not undo itself in half an hour: *normal* reads as *before this file's event*.
- The others: Flow Instability (WELL-00001, 00004), Rapid Productivity Loss (00016, 00020), Scaling in PCK (00024), Hydrate in Service Line (00042).

▸ Appendix

PETROBRAS

Thank you!



github.com/Thiago-TP/3w-well-data-viewer

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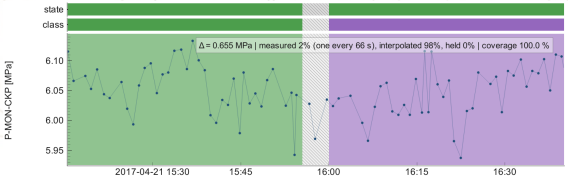
Appendix: label changes across an unlabeled stretch

Timelines page ► How are the instances of one well distributed in time?

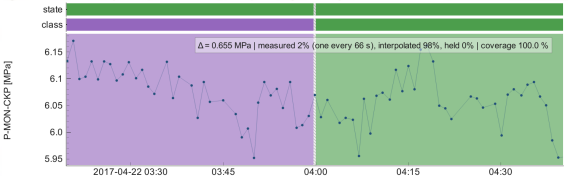
The 16 unlabeled stretches of a joined history between two different labels,

	Well	Unlabeled from	Label before	Label after	For	Shown on
1	WELL-00001	2017-04-21 15:55	normal	Flow Instability (steady)	4 min	P-MON-CKP
2	WELL-00001	2017-04-22 03:59	Flow Instability (steady)	normal	25 s	P-MON-CKP
3	WELL-00001	2017-06-26 09:59	Flow Instability (steady)	normal	3 min	P-MON-CKP
4	WELL-00002	2013-12-12 08:59	normal	Flow Instability (steady)	41 s	P-TPT
5	WELL-00002	2013-12-14 22:00	normal	Flow Instability (steady)	9 s	P-ANULAR
6	WELL-00004	2014-08-06 22:00	Flow Instability (steady)	normal	32 s	P-MON-CKP
7	WELL-00004	2017-10-31 19:14	Quick Restriction in PCK (steady)	normal	5 s	P-MON-CKP
8	WELL-00004	2017-10-31 19:48	Quick Restriction in PCK (steady)	normal	15 s	P-MON-CKP
9	WELL-00006	2018-06-18 11:00	Abrupt Increase of BSW (steady)	normal	30 min	P-MON-CKP
10	WELL-00016	2018-04-26 14:44	Rapid Productivity Loss (transient)	normal	7 min	P-MON-CKP
11	WELL-00020	2014-03-18 01:30	Rapid Productivity Loss (steady)	normal	29 min	P-MON-CKP
12	WELL-00020	2014-03-18 13:57	Rapid Productivity Loss (transient)	normal	53 min	P-MON-CKP
13	WELL-00020	2014-03-19 11:00	Rapid Productivity Loss (transient)	normal	1 h	P-MON-CKP
14	WELL-00024	2016-09-11 07:00	Scaling in PCK (transient)	normal	40 min	P-MON-CKP
15	WELL-00042	2014-12-18 00:53	Hydrate in Service Line (transient)	normal	48 min	P-MON-CKP
16	WELL-00042	2014-12-22 03:37	Hydrate in Service Line (transient)	normal	8 min	P-MON-CKP

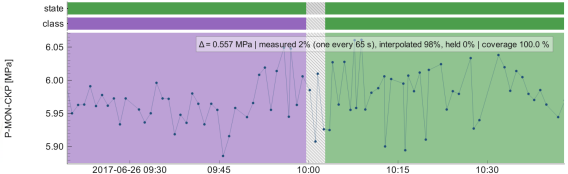
Appendix: label changes 1 to 4



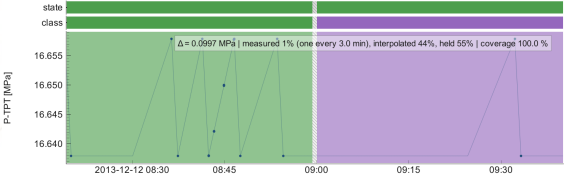
1 WELL-00001, 2017-04-21 15:55: normal → 4 min unlabeled → Flow Instability (steady)



2 WELL-00001, 2017-04-22 03:59: Flow Instability (steady) → 25 s unlabeled → normal

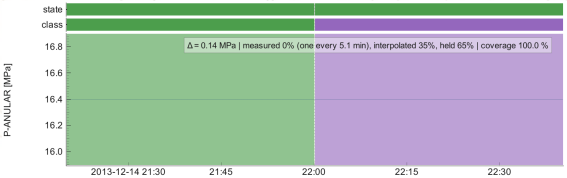


3 WELL-00001, 2017-06-26 09:59: Flow Instability (steady) → 3 min unlabeled → normal

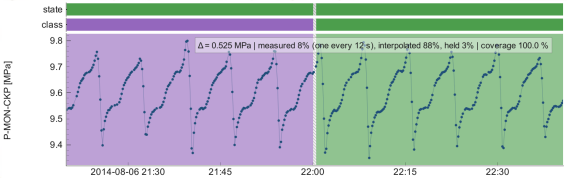


4 WELL-00002, 2013-12-12 08:59: normal → 41 s unlabeled → Flow Instability (steady)

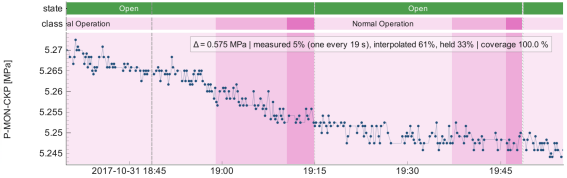
Appendix: label changes 5 to 8



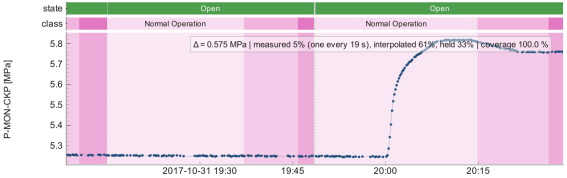
5 WELL-00002, 2013-12-14 22:00: normal → 9 s unlabeled → Flow Instability (steady)



6 WELL-00004, 2014-08-06 22:00: Flow Instability (steady) → 32 s unlabeled → normal

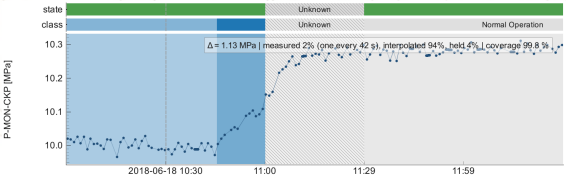


7 WELL-00004, 2017-10-31 19:14: Quick Restriction in PCK (steady) → 5 s unlabeled → normal

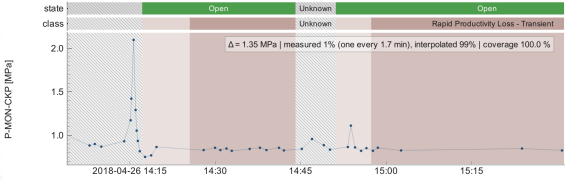


8 WELL-00004, 2017-10-31 19:48: Quick Restriction in PCK (steady) → 15 s unlabeled → normal

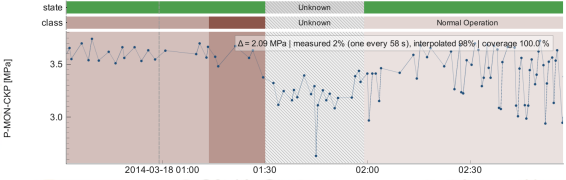
Appendix: label changes 9 to 12



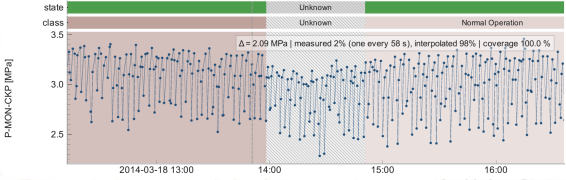
9 WELL-00006, 2018-06-18 11:00: Abrupt Increase of BSW (steady) → 30 min unlabeled → normal



10 WELL-00016, 2018-04-26 14:44: Rapid Productivity Loss (transient) → 7 min unlabeled → normal

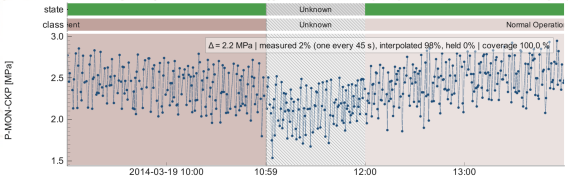


11 WELL-00020, 2014-03-18 01:30: Rapid Productivity Loss (steady) → 29 min unlabeled → normal

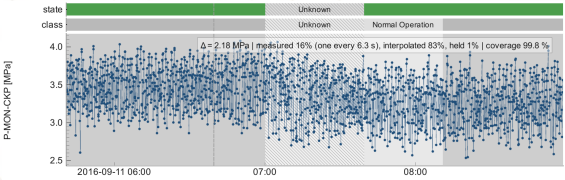


12 WELL-00020, 2014-03-18 13:57: Rapid Productivity Loss (transient) → 53 min unlabeled → normal

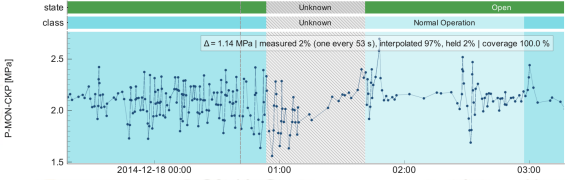
Appendix: label changes 13 to 16



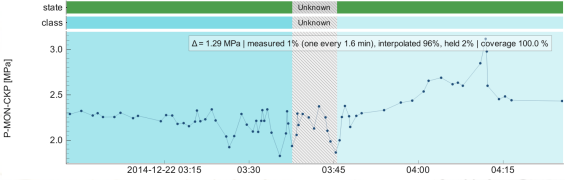
13 WELL-00020, 2014-03-19 11:00: Rapid Productivity Loss (transient) → 1 h unlabeled → normal



14 WELL-00024, 2016-09-11 07:00: Scaling in PCK (transient) → 40 min unlabeled → normal



15 WELL-00042, 2014-12-18 00:53: Hydrate in Service Line (transient) → 48 min unlabeled → normal



16 WELL-00042, 2014-12-22 03:37: Hydrate in Service Line (transient) → 8 min unlabeled → normal